STUDY-RL

Deep Reinforcement Learning

24 core weeks plus applied audit and causal biology

Foundations · Deep RL · LLM RL
Safety-critical RL · Matching · Causal biology

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Preface

This document is the master syllabus for STUDY-RL, a 24-week self-paced curriculum designed for a learner who is fluent in modern deep learning but new to reinforcement learning. An additional applied audit in week 25 connects the core material to a synthetic ATS experiment and an application review. Week 26 adds causal inference and discovery across biological environments, with task-aware enhancement of noisy, sparse measurements. It extends the OPE and offline-data lessons rather than repeating their policy-value estimators. It threads three goals through every week:

1. **Applied mathematics** — connect assumptions, derivations, worked examples, and code; distinguish exact results from approximations.
2. **Modern research topics** — selected readings on DPO/GRPO/RLVR, DreamerV3, EfficientZero v2, BBF, CrossQ, diffusion policies, and LLM agents. Topic coverage is not a claim that every method is implemented or remains the best current choice.
3. **Two applied threads** — *safety-critical RL* (one callout per lecture plus a dedicated 3-week track) and *matching/ranking* (insurance lead matching and recruitment candidate matching, woven through bandits, OPE, fairness).

Each week includes a teaching guide (`teaching_guide.md`), a Tufte-flavored PDF (`notes.pdf`), a Jupyter workbook with graded exercises (`workbook.ipynb`), curated extended reading (`extended_reading.pdf`), and a mini-project with tests (`project/`, `tests/`).

How to use this curriculum

- About **10 hours per week**: 2–3 h reading the PDF, 3–4 h on the workbook, 3–4 h on the project.
- Start with `STUDY_GUIDE.md` in the repository root, then the weekly teaching guide. Predict an update, calculate it by hand, and explain the result before running it.
- Attempt workbook exercises before opening hints or reference solutions. Some supplied cells call the reference directly: these demonstrate a solution but do not test an independent implementation.
- Check both execution and understanding: reproduce a numerical example, explain the assumptions, test a changed input, and compare a project result with a baseline. Passing tests alone does not establish mastery.

1 Curriculum overview

1.1 Tracks

T1 — Foundations (Weeks 1–4) Bandits, MDPs, Bellman equations, dynamic programming, Monte Carlo, TD/SARSA/Q-learning.

T2 — Deep value-based (Weeks 5–6) DQN, Double/Dueling/PER, distributional RL, Rainbow, BBF.

T3 — Policy methods (Weeks 7–10) REINFORCE, A2C, GAE, TRPO, PPO, SAC/TD3, CrossQ.

T4 — Bandits & matching (applied) (Weeks 11–12) LinUCB/TS, OPE, CLTR, fairness, two-sided markets.

T5 — Multi-agent RL (Week 13) MADDPG, MAPPO, self-play, PSRO, AlphaZero case study.

T6 — Offline RL (Weeks 14–15) CQL, IQL, AWR, TD3+BC, Decision Transformer, diffusion policies.

T7 — Model-based RL (Week 16) Dreamer V3, MuZero/EfficientZero V2, TD-MPC2, DIAMOND.

T8 — Exploration & meta-RL (Week 17) RND, ICM, BYOL-Explore, MAML/PEARL, in-context RL.

T9 — LLM RL (Weeks 18–21) RLHF/PPO, DPO/IPO/KTO/ORPO/SimPO, GRPO/RLOO/R-LVR, PRMs and reasoning RL.

T10 — Safety-critical RL (Weeks 22–24) CMDPs, CPO, shielding, formal verification, scalable oversight, capstone.

T11 — Applied audit (Week 25) Synthetic ATS experiments, evaluation assumptions, and an RL enhancement audit.

T12 — Causal biology (Week 26) Confounding, identification versus discovery, Markov equivalence, multi-environment causal learning, and task-aware enhancement.

1.2 Week-by-week schedule

The topic column includes algorithms studied in the readings. The project column describes the bundled implementation; each weekly teaching guide explains its simplifications. A successful toy run is not a reproduction of a paper's benchmark result. Larger benchmark environments require separate experiment configuration and evaluation.

Week	Topic	Bundled mini-project
01	MDPs, bandits, the RL framing	Multi-armed bandit testbed + UCB/TS

Week	Topic	Bundled mini-project
02	Bellman equations, dynamic programming	Value/policy iteration on GridWorld
03	Monte Carlo prediction & control	Blackjack on-policy MC
04	TD, SARSA, Q-learning, Expected SARSA	CliffWalking + Taxi-v3
05	Function approximation & Deep Q-Networks	DQN on LunarLander
06	Rainbow DQN + modern tricks (BBF, NoisyNets)	Rainbow-lite ablations on LunarLander
07	Policy gradients & REINFORCE	REINFORCE + baselines on CartPole
08	A2C, GAE, natural PG, TRPO	A2C+GAE on LunarLander-Continuous
09	Proximal Policy Optimization (deep dive)	PPO on BipedalWalker + the 37 details checklist
10	SAC, TD3, CrossQ for continuous control	SAC on Pendulum
11	Bandits + contextual bandits + OPE	Off-policy evaluation of a synthetic insurance lead–market matcher
12	RL for ranking / matching (counterfactual LTR, fairness, two-sided)	ATS-style candidate–job ranker with fairness audit
13	Multi-agent RL + AlphaZero case study	Tic-Tac-Toe search/self-play and small MAPPO coordination/spread tasks
14	Offline RL fundamentals (CQL, IQL, AWR, TD3+BC)	Tabular IQL, CQL, and naive Q on synthetic GridWorld logs
15	Decision Transformer + diffusion policies	Small DT on synthetic GridWorld trajectories and numerical diffusion exercises
16	Model-based RL (Dreamer V3, MuZero, TD-MPC2)	Dreamer-lite on CartPole
17	Exploration + meta-RL + in-context RL	RND and a full-information MAML-inspired bandit toy
18	RLHF + PPO for LLMs (from scratch and via TRL)	Character-model SFT, reward learning, and PPO; optional real LoRA lab
19	DPO + IPO + KTO + ORPO + SimPO	Character-model preference objectives; optional real DPO lab
20	GRPO + RLOO + RLVR + process reward models	Group-relative updates on synthetic arithmetic prompts
21	Reward hacking, length bias, KL control, scalable oversight	Reward-hack lab + Constitutional-AI mini-loop
22	Constrained MDPs, Lagrangian methods, PPO-Lagrangian, CPO	PPO-Lagrangian on synthetic Lava-GridWorld
23	Safe exploration, shielding, robust RL, formal methods	Model-based grid shield and a pendulum safety toy
24	Capstone: safe candidate-matching policy	End-to-end: offline-pretrained → constrained online → audited
25	Applied synthetic ATS RL enhancement audit	Synthetic ATS logging, OPE, constraints, and audit

Week	Topic	Bundled mini-project
26	Causal discovery across biological environments; task-aware data enhancement	Synthetic patient effects, invariance diagnostics, and held-out-environment enhancement ablations

1.3 Prerequisites

Fluent PyTorch, comfortable with probability, linear algebra, basic optimization theory. You do *not* need prior RL experience.

1.4 Causal-biology extension: one-week plan

Week 26 takes about ten hours across five study sessions. Start after week 11; week 14 supplies additional context about support and extrapolation. The later policy and LLM tracks are not required. Read its teaching guide, calculate a confounded and adjusted effect, compare observationally equivalent graphs, study which mechanisms can change across environments, and evaluate enhancement with patient-disjoint training, validation, and test sets. The workbook and mini-project use invented data with known mechanisms.

The extension is motivated by Synlico's scientific discussion on causal discovery across environments and task-aware enhancement. Primary sources and their limitations are recorded in `research/causal_biology.md`. Multi-environment S-FCI and invariant causal prediction are reading topics; the numerical diagnostic in the project is not a reproduction of either algorithm. Company descriptions of proprietary enhancement do not supply independent evidence of clinical or target-discovery performance.

2 Bibliography pointers

The per-week `extended_reading.pdf` carries the focused paper list for that lecture. A consolidated bibliography is maintained at `research/`.